

NADA MORGAN

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PROFESSIONAL SUMMARY

Mechatronics Engineering student (Ain Shams University and University of East London dual award, graduating 2027) with hands-on electrical design experience: custom one- and two-layer PCBs, buck DC-DC converters, charger IC integration, motor drivers, and protection circuits. Comfortable across the full loop — schematic and layout in EasyEDA, LTspice and Simulink simulation, soldering and bring-up with oscilloscope and multimeter, then debug and redesign.

SKILLS

- **Power Electronics:** buck DC-DC converters, charger IC integration, reverse-polarity and overcurrent protection, power tree design, lithium-ion battery integration
- **PCB Design and Wiring:** EasyEDA schematic capture and layout, one- and two-layer boards, motor-driver and sensor routing, soldering, prototype bring-up and rework, cable management and wire labeling
- **Circuit Simulation:** LTspice, MATLAB Simulink, Simscape, Proteus
- **Lab Instruments:** oscilloscope, multimeter, bench power supply, soldering iron, hot-air rework
- **Motor Drivers and Sensors:** A4988 / DRV8825 stepper drivers, L298 / TB6612 H-bridge DC drivers, IMU (MPU-6050), encoders, IR / proximity sensors, limit switches
- **Embedded Systems:** ESP32, Arduino, Raspberry Pi, RTOS, micro-ROS, I2C, SPI, UART, PWM, ADC
- **Control Systems:** PID (single-loop and cascaded), feedback loops, state machines, SIMATIC S7-1200 PLC, SIL/HIL (Factory I/O)
- **Programming:** C, C++, Python, MATLAB, Ladder Logic

PROJECTS

Battery-Powered Mobile Robot — Two-PCB Power and System Design

Spring 2026

- Designed two custom PCBs in EasyEDA: a power board with two buck DC-DC converters, a charger IC, and reverse-polarity / overcurrent protection; and a main system board carrying the ESP32, DC motor drivers, IMU, and IR sensors.
- Selected and integrated a lithium-ion charger IC with protection circuitry; verified charge/discharge behaviour on the bench before pack integration.
- Brought up both boards: soldered prototypes, probed rails and switching nodes with the oscilloscope, debugged routing issues, and produced a revised PCB spin.
- Developed RTOS firmware on ESP32 (cascaded PID wheel-speed control) with ROS2 navigation on Raspberry Pi over micro-ROS; fused encoder and IMU odometry, validating the IMU in simulation before hardware integration.

Automated Multi-Station Production Line — Drives, Sensors, and Power

Fall 2025

- Specified and wired the electrical subsystem: A4988 / DRV8825 stepper drivers for sorting axes, L298 / TB6612 H-bridge drivers for DC motors, and buck converters stepping station voltage to logic and driver rails.
- Integrated IR proximity sensors and inductive limit switches feeding RTOS firmware across multiple microcontrollers; coordinated inter-station handoffs over ESP-NOW.
- Tested, debugged, and reworked driver circuits and sensor wiring during commissioning; validated state-machine logic in MATLAB Stateflow before build.

PLC-Based Industrial Automation — Liquid Level and Servo Control

Spring 2026

- Programmed a SIMATIC S7-1200 in TIA Portal for a liquid-level loop using PID_Compact; validated via SIL/HIL (Factory I/O against PLCSIM).
- Implemented cascaded position-velocity-current PID on a single-axis servo; modelled the plant in MATLAB Simulink and added a TIA Portal HMI with alarms.

Robotics — Delta Parallel Robot and 4-DOF Arm

2025 – 2026

- Delta robot: derived FK/IK in MATLAB; implemented cascaded PID on Joint 1 (DC motor with encoder feedback) with workspace limits; measured pick-and-place repeatability via position standard deviation.
- 4-DOF arm: modelled in URDF/Xacro, planned collision-aware trajectories in MoveIt 2 / Gazebo, bridged physical joints via micro-ROS on ESP32 for pick-and-place sync with RViz.

EXPERIENCE

Aerodynamics Engineer — ASU Racing Team, Formula Student UK

Aug 2024 – Present

- Designed the FSUK 2025 rear wing (4.5% downforce-to-drag improvement via ANSYS Fluent CFD); contributed to a 3rd overall finish. Now researching closed-loop embedded sensor/actuator integration for active aerodynamic devices.

EDUCATION

B.Sc. Mechatronics Engineering — Ain Shams University / University of East London (Dual Award)

Sep 2023 – 2027

Coursework: control systems, embedded firmware, industrial automation, power electronics, robotics. IGCSE: A in A-Level Mathematics; 9 A* at O-Level. Languages: Arabic (native), English (fluent).